

LESSON 8.9

SOLVING SYSTEMS BY GRAPHING



LESSON INTRODUCTION

MA.8.AR.4.3 Given a mathematical or real-world context, solve systems of two linear equations by graphing.

LEARNING TARGETS

- I can solve a system of linear equations by graphing.
- I can verify my answer from graphing by substituting in the x- and y-coordinates of the solution(s).

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two variable linear equations algebraically or graphically.

ESSENTIAL QUESTIONS

- How do you approximate the solution to a system of equations from a graph?
- How many solutions do systems of equations have?
- How can you determine the number of solutions a system of equations has?

LESSON NARRATIVE

This lesson focuses on students solving systems of equations by graphing and verifying the solutions using substitution. Students will also learn how to approximate solutions to systems when the point of intersection includes non-integer coordinate values. Students will also further explore a system of parallel lines to determine the number of solutions and characteristics. By the end of the lesson, students will be able to summarize the three types of solutions and the characteristics of the systems that match those solution types.

COMPONENT SUMMARY

8.9.1 WARM-UP (~5 MIN.)

Choose and justify which graph does not belong with the others

8.9.3 GUIDED INSTRUCTION (10-15 MIN.)

Approximate solutions to systems of equations from graphs

8.9.5 BRING IT TOGETHER (5 MIN.)

Summarize features of systems with different numbers of solutions

8.9.2 COLLABORATION (10 MIN.)

Solve systems of linear equations by graphing, and verify answers algebraically

8.9.4 EXPLORATION (10 MIN.)

Explore a system of parallel lines

8.9.6 WRAP-UP (~5 MIN.)

Solve a system of linear equations by graphing

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- System of equations
- Parallel lines

MATERIALS

- (Optional) Scientific calculator

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may mix up the x- and y- values when verifying the solution found graphically.
- Students may approximate solutions incorrectly due to their lack of numeracy.
- Students may graph the equations incorrectly leading to incorrect solutions.

THINGS TO CONSIDER

- When students approximate solutions, it may be helpful to have students compare approximations and their verifications with someone else to be able to discuss how they can tell if their approximation is reasonable.

LITERACY AND LANGUAGE ROUTINES

Take Turns

8.9.2 Collaboration positions students to work with a partner to find solutions to systems by graphing and verify their answers using substitution. Consider using the “Take Turns” Language Routine to encourage students to partake in discussions where they alternate explaining their reasoning and listening to the reasoning of others.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

This lesson includes multiple opportunities for students to discuss and communicate mathematical ideas, vocabulary, and methods effectively. Students will be able to analyze the mathematical thinking of others and construct possible arguments based on evidence as they continue to explore systems of equations.

DIFFERENTIATE INSTRUCTION

Consider how to use data from Lessons 7 and 8 to determine partners and possible small groups to pull for differentiated support in the 8.9.2 Collaboration.

Additional differentiation opportunities are denoted throughout the Lesson Guide.

8.9.1 WARM-UP: WHICH ONE DOESN'T BELONG?

TEACHER MOVES

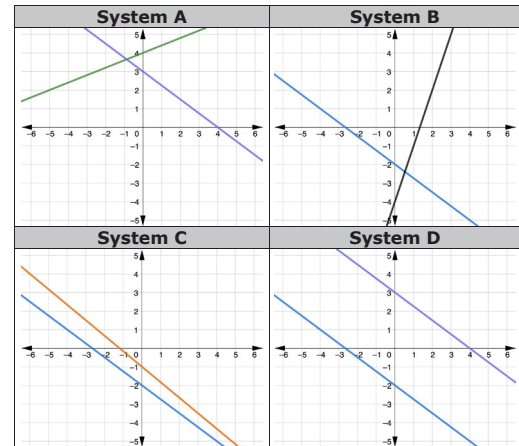
Which One Doesn't Belong?

If time allows, consider providing students with the opportunity to share their answer and explanation with a partner. Then, give them a chance to revise their rationale based on feedback from their partner. This strategy fosters a need to define terms carefully and use words precisely in order to compare and contrast the group of graphs.



8.9.1 Warm-Up: Which One Doesn't Belong?

1. The graphs of four systems of equations are shown.



- a. Which graph does not belong?

Answers will vary.

- b. Explain your reasoning.

Answers will vary.

System A does not belong because it is the only system with a solution in Quadrant II.

System C does not belong because it is the only one with two lines that have negative slopes that intersect.

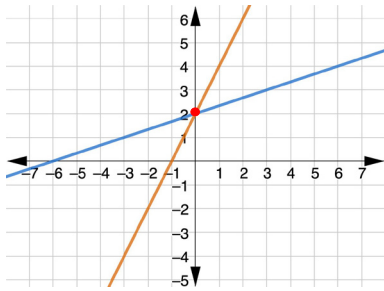
System D does not belong because the two lines are parallel and will never intersect.



8.9.2 Collaboration: Solving Systems from Graphs

Work with your partner to complete the following.

1. The graph of the system $\begin{cases} y = \frac{1}{3}x + 2 \\ y = 2x + 2 \end{cases}$ is shown.



- Mark the solution to the system on the graph.
Shown on graph.
- What are the coordinates of the solution to the system?
(0, 2)
- Verify the solution using substitution.

$y = \frac{1}{3}x + 2$	$y = 2x + 2$
$2 = \frac{1}{3}(0) + 2$	$2 = 2(0) + 2$
$2 = 0 + 2$	$2 = 0 + 2$
$2 = 2$	$2 = 2$

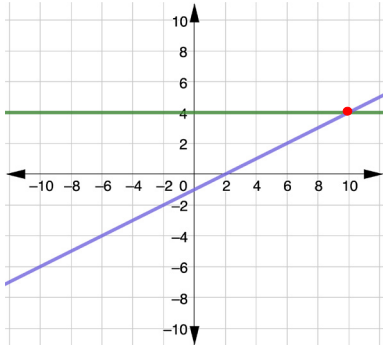
8.9.2 COLLABORATION: SOLVING SYSTEMS FROM GRAPHS

DIFFERENTIATE INSTRUCTION

Consider using information from the previous two lessons to form a small group of students who are struggling working with systems of equations. Facilitate these students as they work through 8.9.2 Collaboration.

Strategically pair the other students for success based on observations over the last two lessons.

2. The graph of the system $\begin{cases} y = 4 \\ y = \frac{1}{2}x - 1 \end{cases}$ is shown.



- Mark the solution to the system on the graph.
Shown on graph.
- What are the coordinates of the solution to the system?
(10, 4)
- Verify the solution using substitution.
 $y = 4$
 $4 = 4$
 $y = \frac{1}{2}x - 1$
 $4 = \frac{1}{2}(10) - 1$
 $4 = 5 - 1$
 $4 = 4$

The solution satisfies both equations.

TEACHER MOVES

Take Turns

Consider using the “Take Turns” language routine while students work in partners to complete this component of the lesson. This routine encourages students to explain their logic as well as listen to and analyze the reasoning of others.

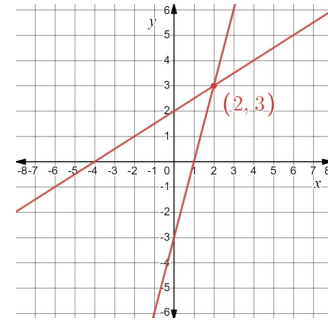
8.9.2 COLLABORATION: SOLVING SYSTEMS FROM GRAPHS (CONTINUED)

TEACHER MOVES

Students need to use a straightedge to create their graphs.

Students graphed equations in Unit 7, so all students should not need a refresher. However, if data from the prior unit indicates that students may still be struggling with graphing, the teacher might consider providing additional support/scaffolding for students who still need it at this point.

3. Use the coordinate plane to complete the following.



- Graph $y = 3x - 3$ on the coordinate plane.
Shown on graph.
- Graph $y = \frac{1}{2}x + 2$ on the coordinate plane.
Shown on graph.
- Mark and label the solution to the system $\begin{cases} y = 3x - 3 \\ y = \frac{1}{2}x + 2 \end{cases}$ on your graph.
Shown on graph.
- Verify your solution using substitution.

$$\begin{aligned} y &= 3x - 3 \\ 3 &= 3(2) - 3 \\ 3 &= 6 - 3 \\ 3 &= 3 \end{aligned}$$

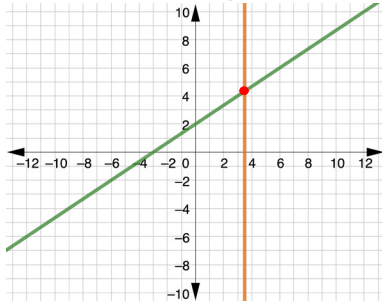
$$\begin{aligned} y &= \frac{1}{2}x + 2 \\ 3 &= \frac{1}{2}(2) + 2 \\ 3 &= 1 + 2 \\ 3 &= 3 \end{aligned}$$

The solution satisfies both equations.



8.9.3 Guided Instruction: Approximating Solutions

1. The graph of the system $\begin{cases} x = 3.5 \\ y = \frac{2}{3}x + 2 \end{cases}$ is shown.



- a. Mark the solution to the system of equations.
Shown on graph.

The solution to the system does not fall on the grid lines shown on the coordinate grid. Therefore, the solution can only be approximated from the graph.

- b. What are the approximate coordinates of the solution?
Answers will vary. (3.5, 4.5)
- c. Kyrie said, "I know the x -value of the solution is 3.5. I could use that to determine the y -value of the solution." Discuss with your partner what Kyrie may have been thinking.
Answers will vary. Because one of the equations is $x = 3.5$, Kyrie may have been thinking to substitute 3.5 for x in the other equation to find the y -value.

8.9.3 GUIDED INSTRUCTION: APPROXIMATING SOLUTIONS

TEACHER MOVES

Explain that sometimes the exact solution to a system of equations cannot be determined from the graph. Model how to approximate the solution to a system of equations from a graph using Think-Aloud strategies.

Poll students for their approximations on question B, pushing not only for their approximation, but how they determined it. This could also be done by asking students to write their approximation and then generating a list of answers observed on the board to discuss and narrow down the best approximation. This provides an opportunity to highlight student thinking, including errors in reasoning, without assigning any of the approximations to any one student.

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

Listen as students discuss their thinking. Consider asking the following:

- Why do you think Kyrie said he knows the x -value of the solution is 3.5?
- If Kyrie knows the x -value, how do you think he could use that to determine the y -value?

8.9.3 GUIDED INSTRUCTION: APPROXIMATING SOLUTIONS (CONTINUED)

TEACHER MOVES

Depending on student discussions and understanding in question 2c, consider releasing students to test Kyrie's conjecture with a partner or modeling how to do so while thinking aloud.

DIFFERENTIATE INSTRUCTION

Allow informal observation so far in the lesson to dictate how "guided" question 2 should be for you and your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher modeling with students actively working as a whole group.

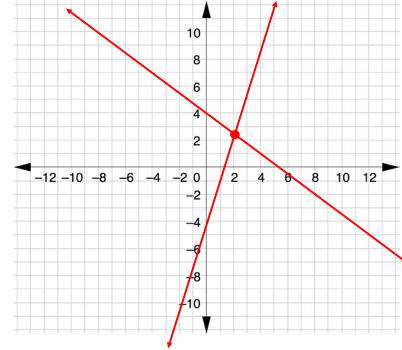
TEACHER MOVES

Facilitate a discussion to clarify that only the approximate solution can be determined from the graph. Explain that students will learn how to find the exact solution algebraically in Algebra 1.

- d. Test Kyrie's conjecture to see if his strategy will work to determine the exact solution of the system.

$$\begin{aligned} y &= \frac{2}{3}x + 2 \\ y &= \frac{2}{3}(3.5) + 2 \\ y &= \frac{7}{3} + 2 \\ y &= \frac{13}{3} \end{aligned}$$

2. Use the coordinate plane to complete the following.

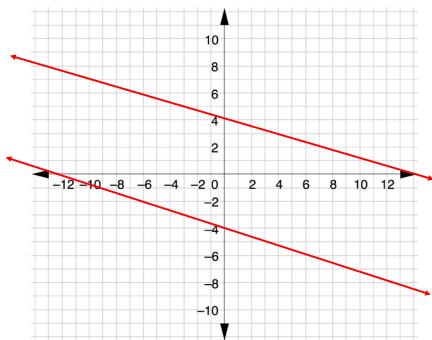


- a. Graph the system $\begin{cases} y = -\frac{3}{4}x + 4 \\ y = 3x - 4 \end{cases}$ on the coordinate plane.
Shown on graph.
- b. Mark the solution to the system on your graph.
Shown on graph.
- c. What are the approximate coordinates of the solution?
(2.1, 2.4)



8.9.4 Exploration: Relationships in Parallel Lines

- Use the coordinate grid below to complete the following.



- Graph the system
$$\begin{cases} y = -\frac{1}{3}x + 4 \\ y = -\frac{1}{3}x - 4 \end{cases}$$
.

Shown on graph.

- Compare your graph with your partner and resolve any differences, if necessary.
- Discuss with your partner the solution(s) to the system based on your graphs.
Answers will vary. The system will have no solutions because the lines are parallel. There is no point that satisfies both equations because they will never intersect.

- Work with your partner to determine which statements are true for this system. Then, check all the true statements.
 - ☐ The two lines are perpendicular.
 - ☐ *There is no solution to the system.*
 - ☐ *There are no points that lie on the graph of both lines.*
 - ☐ There are infinitely many solutions to the system.
 - ☐ Any point will make both equations true.
 - ☐ *The two lines are parallel.*

- Compare the two equations in the system. Discuss with your partner what is the same and what is different about the equations.

- Complete the statements.

A system of equations that has no solutions occurs

when the lines are
☐ parallel.
☐ perpendicular.
 The

equations of
☐ parallel
☐ perpendicular
 lines have the

same
☐ slopes,
☐ y-intercepts,
 but different

☐ slopes.
☐ y-intercepts.

8.9.4 EXPLORATION: RELATIONSHIPS IN PARALLEL LINES

TEACHER MOVES

Consider providing students with sentence stems to support rich discussion when comparing their graphs.

- I started the graph of the first line by...
- I started the graph of the second line by...
- I agree with you because...
- I disagree with you because...
- One thing I noticed was ...

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

Listen as students discuss their thinking in response to this prompt to highlight student thinking and/or to address misconceptions.

TEACHER MOVES

Monitor while students are working to determine if there are any questions that need to be debriefed to start the 8.9.5 Bring It Together. This could include highlighting student voices to share rich discussions, comments overheard while they collaborated for the whole class to hear, or to address common misconceptions that were evident among the majority of the class.

8.9.5 BRING IT TOGETHER: SOLUTIONS TO SYSTEMS OF EQUATIONS

TEACHER MOVES

This component is intended to be teacher-led. Consider first allowing students to take turns reading the descriptions of the systems with their partners and discussing their initial thinking. If student discussion indicates that students are making the necessary connections, ask them to complete question 1 with their partner.

When going over question 2, consider pointing to specific systems students have explored in this lesson and Lesson 8 to make connections between the number of solutions and the features of the equations in the system.

8.9.5 Bring It Together: Solutions to Systems of Equations

1. The table gives three descriptions of the types of solutions to systems of linear equations. Match each description to the number of solutions the system has.

Description of the System	Number of Solutions
All the ordered pairs that satisfy one equation will also satisfy the other equation.	☐ one solution ☐ no solutions ☐ infinitely many solutions
Exactly one ordered pair will satisfy both equations. It is the point of intersection of the graphs.	☐ one solution ☐ no solutions ☐ infinitely many solutions
There are no ordered pairs that will satisfy both equations.	☐ one solution ☐ no solutions ☐ infinitely many solutions

2. The features of the equations in different types of systems of two linear equations are given. Match each description to the number of solutions the system has.

Description of the System	Number of Solutions
The two lines have different slopes.	☐ one solution ☐ no solutions ☐ infinitely many solutions
The two lines have the same slope and the same y-intercept.	☐ one solution ☐ no solutions ☐ infinitely many solutions
The two lines have the same slope and the different y-intercepts.	☐ one solution ☐ no solutions ☐ infinitely many solutions



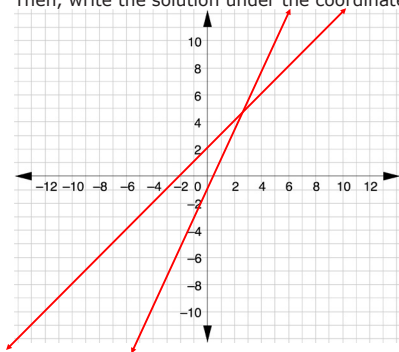
8.9.6 Wrap-Up: Ticket Out the Door

1. Two systems are shown.

$$\begin{cases} y = x + 2 \\ y = 2x - 1 \end{cases}$$

$$\begin{cases} y = -x + 3 \\ y = -x - 1 \end{cases}$$

- Circle the system that will have no solutions.
- Explain how you know it will have no solutions.
Answers will vary. The system will have no solutions because the slopes are the same and when the slopes are the same and the y-intercepts are different, the system will have no solutions.
- Graph the system you did NOT circle in Part A. Then, write the solution under the coordinate plane.



Solution (3 , 5)

8.9.6 WRAP-UP: TICKET OUT THE DOOR

ENRICHMENT

Consider asking students to compare and contrast a system of equations with no solution and a system of equations with infinitely many solutions. How are they the same or different algebraically? Graphically?